

Positive-Density Two-Sided Product Escape for Frequently Hypercyclic Bilateral Shifts

Candidate partial result for arXiv:1305.2325

13 August 2026

Status: candidate partial result, pending human review. For every invertible frequently hypercyclic bilateral weighted shift on $c_0(\mathbb{Z})$, both one-sided cumulative products escape in the required directions along one set of positive lower density. A tail-richness condition on the standard witness family then implies that the inverse is frequently hypercyclic. The unrestricted bilateral-shift inverse problem remains open.

1 The surviving problem

Let $w = (w_n)_{n \in \mathbb{Z}}$ be a bounded and bounded-below sequence of positive weights and let $B_w e_n = w_n e_{n-1}$ on $c_0(\mathbb{Z})$. Put

$$W_r = \prod_{\nu=1}^r w_\nu, \quad L_r = \prod_{\nu=-r+1}^0 w_\nu \quad (r \geq 1). \quad (1)$$

Bayart and Ruzsa ask whether frequent hypercyclicity of an invertible B_w forces frequent hypercyclicity of B_w^{-1} [1]. Their PDF contains the evident typo “is T^{-1} invertible?”; the next paragraph and their partial theorem make the intended question unambiguous.

We now turn to an interesting problem regarding frequently hypercyclic operators.

Question. Let $T \in \mathcal{L}(X)$ be frequently hypercyclic and invertible. Is T^{-1} invertible?

In view of this paper, it is natural to study whether a bilateral weighted shift on c_0 could be a counterexample. If we look at Theorem [12](#), this does not seem impossible; indeed, because of (c), the conditions on the right part and on the left part of B_w are not symmetric. However, it could be possible that this condition is superfluous.

Figure 1: The question and bilateral-shift specialization in arXiv:1305.2325, PDF page 16.

Menet later answered the question negatively for general operators on ℓ^1 [3]. The special bilateral-shift problem, however, is explicitly retained in Grosse-Erdmann’s later characterization [2].

This extends the case of frequent hypercyclicity in [4 Theorem 12] to not necessarily invertible operators and to arbitrary Furstenberg families.

But the question posed by Bayart and Ruzsa [4 p. 707] remains: can one drop the (non-symmetric) condition (ii) for frequent hypercyclicity (as we did in the previous section for upper Furstenberg families)? If so then the bilateral weighted shifts on $c_0(\mathbb{Z})$ could not serve as counter-examples to Question [1]

Figure 2: The surviving non-symmetric condition in arXiv:1707.03994, PDF page 13.

We use the following invertible specialization of that characterization. For every null sequence $\varepsilon_p > 0$, frequent hypercyclicity of B_w provides pairwise disjoint sets $E_p \subset \mathbb{N}$ with $\underline{d}(E_p) > 0$ such that

$$|n - m| > p + q \quad (n \in E_p, m \in E_q, n \neq m), \quad (2)$$

$$W_n \rightarrow \infty \quad (n \rightarrow \infty, n \in E_p), \quad (3)$$

and, writing $r = |n - m|$,

$$W_r > \frac{1}{\min(\varepsilon_p, \varepsilon_q)}, \quad L_r < \min(\varepsilon_p, \varepsilon_q). \quad (4)$$

The published condition states one inequality for each ordering of n, m ; swapping the ordered pair gives the symmetric formulation (4).

The inverse is conjugate to the backward shift with $w'_n = 1/w_{-n+1}$. Its cumulative products are

$$W'_r = L_r^{-1}, \quad L'_r = W_r^{-1}. \quad (5)$$

Thus (2) and (4) are already invariant under inversion. Only the individual condition (3) is missing on the inverse side.

2 An unconditional positive-density escape theorem

We first record a density diagonalization lemma.

Lemma 1 (lower-density diagonal). *Let $S_1 \supset S_2 \supset \dots$ be subsets of $\mathbb{N}$ and suppose $\underline{d}(S_j) \geq \delta > 0$ for every j . There is $A \subset \mathbb{N}$ with $\underline{d}(A) \geq \delta$ such that $A \setminus S_j$ is finite for every j .*

Proof. Choose increasing integers N_j so that, whenever $N \geq N_j$,

$$\frac{|S_j \cap [0, N]|}{N+1} \geq \delta - \frac{1}{j}.$$

Let $q(n)$ be a nondecreasing integer-valued function tending to infinity such that $N_{q(n)} \leq n$ (define it arbitrarily before N_1), and set

$$A = \{n : n \in S_{q(n)}\}.$$

For $n \leq N$, one has $q(n) \leq q(N)$ and hence $S_{q(N)} \subset S_{q(n)}$. Therefore

$$|A \cap [0, N]| \geq |S_{q(N)} \cap [0, N]| \geq \left(\delta - \frac{1}{q(N)} \right) (N+1).$$

This proves the density bound. Once $q(n) \geq j$, membership in A implies membership in $S_{q(n)} \subset S_j$, proving the almost-containment. $\square$

Theorem 2 (simultaneous two-sided escape). *If B_w is invertible and frequently hypercyclic on $c_0(\mathbb{Z})$, then there is $\delta > 0$ such that, for every $R \geq 1$,*

$$\underline{d}\{r \geq 1 : W_r > R \text{ and } L_r < R^{-1}\} \geq \delta. \quad (6)$$

Consequently there is a set $A \subset \mathbb{N}$ with $\underline{d}(A) \geq \delta$ such that

$$W_r \longrightarrow \infty, \quad L_r \longrightarrow 0 \quad (r \rightarrow \infty, r \in A). \quad (7)$$

Proof. Take a witness family satisfying (2)–(4), and put $\delta = \underline{d}(E_1) > 0$. Given R , choose q so large that $\min(\varepsilon_1, \varepsilon_q) < R^{-1}$, and fix $t \in E_q$. If $n \in E_1$ and $n > t$, then applying (4) to n, t gives

$$W_{n-t} > R, \quad L_{n-t} < R^{-1}.$$

The positive part of the translate $E_1 - t$ has lower density δ , which proves (6). Apply Lemma 1 to the decreasing sets

$$S_j = \{r : W_r > j, L_r < 1/j\}$$

to obtain (7). $\square$

This rules out the most direct counterexample strategy: the inverse cannot fail merely because its positive cumulative products lack a positive-density escape sequence. What is not controlled is the product behavior on all *pairwise differences* of a full family of return-time sets.

3 A tail-rich criterion that gives the full inverse conclusion

Call a witness family (E_p) in (2)–(4) *tail-rich* if there exists $A \subset \bigcup_p E_p$ with $\underline{d}(A) > 0$ such that

$$\ell(n) \longrightarrow \infty \quad (n \rightarrow \infty, n \in A), \quad (8)$$

where $\ell(n) = p$ for $n \in E_p$. The label is well-defined because the E_p are disjoint.

Theorem 3 (tail-rich inversion). *If an invertible frequently hypercyclic bilateral shift B_w admits a tail-rich witness family, then B_w^{-1} is frequently hypercyclic.*

Proof. Fix s and $t \in E_s$, and set

$$B = (A - t) \cap \mathbb{N}.$$

Then $\underline{d}(B) = \underline{d}(A) > 0$. Partition B into sets B_j of positive lower density: if $B = \{b_0 < b_1 < \dots\}$, one may assign b_k to B_j when the 2-adic valuation of $k + 1$ is $j - 1$. Indeed,

$$|B_j \cap [0, N]| = 2^{-j} |B \cap [0, N]| + O(1). \quad (9)$$

Let $\eta_j = 2^{-j}$. Since $\varepsilon_p \rightarrow 0$, choose $P_j \geq j$ such that $\varepsilon_p \leq \eta_j$ for $p \geq P_j$. By (8), deleting finitely many points from each B_j gives sets F_j of positive lower density such that

$$x + t \in E_p, x \in F_j \implies p \geq P_j. \quad (10)$$

For distinct $x \in F_j$ and $y \in F_k$, write $x + t \in E_p$ and $y + t \in E_q$. Translation preserves differences, so (2), (4), and (10) give

$$\begin{aligned} |x - y| &> p + q \geq j + k, \\ L_{|x-y|} &< \min(\eta_j, \eta_k), \\ W_{|x-y|} &> 1/\min(\eta_j, \eta_k). \end{aligned}$$

By (5), these are exactly the separation and pairwise-product conditions for B_w^{-1} .

It remains to check the individual non-symmetric condition. If $x \in F_j$ and $x + t \in E_p$, then (4), applied to $x + t$ and the fixed point $t \in E_s$, gives

$$L_x < \min(\varepsilon_p, \varepsilon_s) \leq \varepsilon_p.$$

As $x \rightarrow \infty$ in F_j , (8) gives $p \rightarrow \infty$, hence $L_x \rightarrow 0$ and $W'_x = L_x^{-1} \rightarrow \infty$. Grosse-Erdmann's characterization applied to (F_j) and (η_j) proves that B_w^{-1} is frequently hypercyclic. $\square$

The hypothesis has a convenient sufficient form.

Corollary 4 (tail-union test). *For a witness family put $U_P = \bigcup_{p \geq P} E_p$. If*

$$\inf_{P \geq 1} d(U_P) > 0, \tag{11}$$

then B_w^{-1} is frequently hypercyclic.

Proof. Apply Lemma 1 to the decreasing sets U_P . The resulting positive-density set is almost contained in every U_P , so its label tends to infinity. Theorem 3 applies. $\square$

Corollary 5 (monotone negative side). *If L_r is eventually nonincreasing (in particular, if $w_n \leq 1$ for all sufficiently negative n), then B_w^{-1} is frequently hypercyclic.*

Proof. Theorem 2 and eventual monotonicity imply $L_r \rightarrow 0$ on all of $\mathbb{N}$. Hence the original sets E_p satisfy the missing individual condition $W'_n = L_n^{-1} \rightarrow \infty$; (2), (4), and (5) provide every other condition. $\square$

4 Remaining obstruction

Condition (11) is not automatic for a countable disjoint family of positive-lower-density sets: their densities may decrease so fast that the tail unions have lower density tending to zero. Theorem 2 does give uniformly dense product sublevel sets, but they contain different common translates of the witness family. Splicing those translates destroys the pairwise-difference control in (4), replacing a controlled difference by a four-term expression. Conversely, a hierarchical counterexample must respect (6)–(7), so merely making the negative products small on successively sparser levels cannot work.

Eight focused routes were audited: exact literature status, symmetric product reformulation, uniform-density extraction, density diagonalization, direct use of one escape set, common-translation gluing, tail-union forcing, and a hierarchical counterexample design. The successful deductions are Theorems 2 and 3; the unrestricted gluing step remains open.

References

- [1] F. Bayart and I. Z. Ruzsa, *Difference sets and frequently hypercyclic weighted shifts*, Ergodic Theory Dynam. Systems 35 (2015), 691–709; arXiv:1305.2325.
- [2] K.-G. Grosse-Erdmann, *Frequently hypercyclic bilateral shifts*, Glasgow Math. J. 61 (2019), 271–286; arXiv:1707.03994.
- [3] Q. Menet, *Inverse of frequently hypercyclic operators*, J. Inst. Math. Jussieu 21 (2022), 1867–1876; arXiv:1910.04452.